

MoodleQuest: Gamified Engagement for Asynchronous Learning

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ABSTRACT

MoodleQuest is a gamified companion system designed to improve student participation in asynchronous Moodle courses while preserving transparency and institutional trust. Prior work shows that timely engagement cues, visible rewards, and structured incentives can strengthen motivation and course persistence [1–3]. Studies on online learning also highlight how live or near-real-time feedback supports higher completion rates [4], while multi-factor analytics can reveal gaps in students' cognitive, social, and emotional engagement [5]. Guided by these insights, MoodleQuest combines quests, experience points, badges, leaderboards, virtual pets, and clear progress indicators to encourage steady study habits and more active interaction with course material.

The system links Moodle through secure single sign-on, role-based access control, and web-service integration, allowing data synchronization without creating duplicate accounts. A Next.js interface and FastAPI middleware deliver dashboards that visualize activity heatmaps, streak patterns, and early warnings for at-risk students. Performance targets include sub-five-second load times and stable operation with more than one hundred concurrent users. Initial walkthroughs covering authentication, quest assignments, badge awarding, leaderboard updates, and progress monitoring suggest that daily quests and streak-based rewards help students better understand their effort and maintain learning momentum.

Planned pilot tests will evaluate teacher workload, student sentiment, and the clarity of privacy notices to refine pacing, prompts, and tracking settings. Identified risks include dependence on teacher validation, uneven internet access, and the phased deployment of deeper analytics. Future development will focus on classroom pilots, privacy-aware telemetry, and recommendation tools that align quests with course objectives. Overall, MoodleQuest offers a practical approach to layering trusted gamification onto Moodle while maintaining visible control for both students and instructors.

1. INTRODUCTION

Asynchronous online learning has become a central mode of instruction in many schools and universities, yet students often struggle with motivation, sustained participation, and meaningful engagement when learning independently. While platforms such as Moodle provide structured content delivery, assignment workflows, and basic activity logs, they offer limited mechanisms that support continuous motivation or provide clear indicators of learning effort. Without real-time feedback or interactive cues, students may treat coursework passively, leading to irregular study habits, reduced retention, and difficulty in measuring their own progress.

Gamification has emerged as a promising approach to address these issues. Research shows that visible rewards, timely engagement cues, and structured interactive elements can reinforce learning behaviors and improve completion rates in digital courses. However, implementing gamification directly within an LMS often requires heavy customization, which may create privacy concerns, administrative overhead, or disruptions to existing workflows.

MoodleQuest was developed to address these gaps by serving as a companion system that overlays gamified engagement and learning validation on top of Moodle through secure integration. Rather than replacing existing workflows, it adds quests, badges, experience points, leaderboards, virtual pets, and analytics that help students monitor their effort and progress. The system aims to provide instructors with clearer engagement insights while ensuring data transparency and user control. This paper presents the design, implementation, and evaluation plan for MoodleQuest as a practical and privacy-aware way to support motivation in asynchronous courses.

2. RELATED WORK

Prior research emphasizes the difficulty of sustaining engagement in asynchronous online courses. Wakjira and Bhattacharya [1] demonstrated that visual engagement indicators help instructors identify disengaged learners early. Kim et al. [3] showed that clustering techniques applied to activity data can more accurately characterize student behavior than basic participation counts. Ang et al. [2], using the Community of Inquiry framework, highlighted

the importance of structured interaction to support cognitive and social presence in asynchronous environments.

Other studies focus on feedback and analytics. Porter and Bozkaya [4] found that frequent or near-real-time feedback is associated with improved completion rates and learning outcomes. Doboli et al. [5] further demonstrated that multi-dimensional analytics incorporating cognitive, social, and emotional indicators can expose learning gaps not visible in standard LMS logs.

Gamification research aligns with these findings, showing that experience points, achievements, and challenge-based activities can enhance persistence and learner motivation. However, many existing systems rely on deep LMS modification or opaque AI-driven mechanisms with limited instructor oversight. MoodleQuest builds on prior work by emphasizing transparent, teacher-guided gamification layered beside Moodle rather than embedded within it.

3. PROBLEM STATEMENT

Although Moodle effectively manages content and assessments, it lacks mechanisms that sustain motivation and reveal real engagement in asynchronous settings. This creates several challenges:

1. **Passive participation** – Students may log in, open materials, or submit work without actively engaging or studying, making activity logs unreliable indicators of effort.
2. **Limited feedback and motivation** – Asynchronous learning often lacks timely incentives or cues that help students maintain consistent study habits.
3. **Insufficient visibility for teachers** – Educators have difficulty identifying disengaged students early because available analytics are minimal and do not capture behavioral patterns.
4. **Lack of integrated gamification** – Moodle provides limited built-in gamification tools and customizing it may require technical expertise or compromise platform stability.
5. **Privacy and trust concerns** – Heavy tracking or automated scoring systems can create discomfort among students and increase administrative burdens.

MoodleQuest addresses these issues by providing a companion system that adds gamified engagement, clear progress indicators, and teacher-guided monitoring while maintaining privacy and seamless Moodle integration.

4. SYSTEM DESIGN AND ARCHITECTURE

MoodleQuest follows a modular, two-tier architecture consisting of (1) a front-end engagement layer and (2) a middleware bridge that communicates with Moodle’s web services:

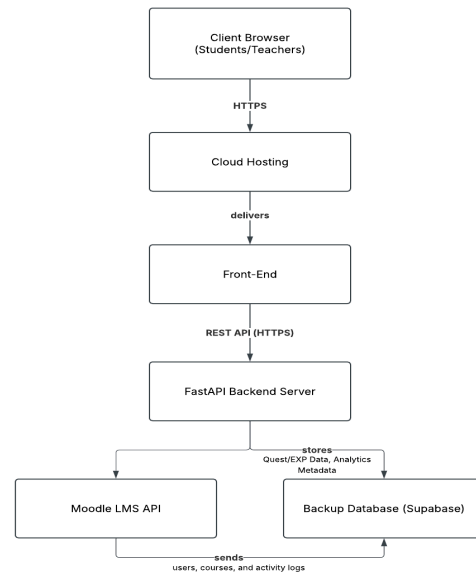


Figure 1. Deployment Architecture Diagram

4.1 Front-End Layer

The user interface is built with NextJS and TailwindCSS, providing responsive dashboards accessible on both mobile and desktop. Features include:

- Gamification dashboard (EXP, badges, virtual pet status)
- Quest and mission manager
- Leaderboards with filters and search
- Engagement heatmap and streak tracker
- Student and teacher analytics views

4.2 Middleware Layer

A FastAPI service acts as a secure bridge between the front-end and Moodle:

- Handles authentication and session verification
- Uses role-based access control to manage student/teacher/admin permissions
- Fetches and updates Moodle data via REST web services
- Stores supplementary data (e.g., quest logs, streaks, badge events) in a backup database

4.3 Integration with Moodle

MoodleQuest does not alter Moodle’s internal structure. Instead, it:

- Uses Moodle SSO for login
- Retrieves course data, activity completion, and grades through official APIs
- Syncs progress and engagement metrics without requiring duplicate accounts
- Respects Moodle’s permissions model to prevent unauthorized access

4.4 Non-Functional Considerations

- Expected capacity: 100+ concurrent users
- Dashboard response target: under 5 seconds
- Encryption: TLS + Moodle’s built-in hashing mechanisms

- Privacy: granular tracking controls and explicit consent prompts

5. IMPLEMENTATION

Implementation followed an agile, sprint-based workflow with four main development phases:

Sprint 1 – Authentication and User Management

- Integration of Moodle SSO via token-based API
- Role-based access control for students, teachers, and administrators
- User profile synchronization and secure session handling

Sprint 2 – Core Gamification Features

- EXP engine with configurable reward rules
- Badge and achievement system
- Virtual pet evolution logic based on engagement
- Leaderboards with daily, weekly, and all-time filters

Sprint 3 – Quests and Progress Analytics

- Teacher tools for creating and assigning quests tied to learning objectives
- Daily quests and streak tracking
- Engagement heatmaps visualizing hourly and weekly activity
- Early-warning flags for low-activity students

Sprint 4 – Optimization and Testing

- Load testing for concurrency targets
- Refining chart rendering and caching for faster performance
- Usability walkthroughs with teachers for adjusting terminology, pacing, and privacy prompts
- Preparation for pilot deployment in a controlled classroom environment

6. SURVEY METHODOLOGY AND DATA COLLECTION

6.1 Participants

A total of **32** students from Bankal National High School participated in the survey. All participants had prior experience using the MoodleQuest system during the pilot testing phase.

6.2 Survey Instrument

The evaluation employed a survey instrument consisting of 15 questions divided into two sections. The quantitative section included **12 Likert-scale** items designed to measure three constructs: Perceived Engagement, Usability, and Learning Impact. The qualitative section consisted of three open-ended questions that gathered feedback on students' experiences, perceived difficulties, and suggested improvements to the system.

6.3 Measurement Scale

All quantitative items used a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). Higher scores indicate a more positive perception of MoodleQuest. This scale was applied consistently across all quantitative items to allow comparison among the evaluated constructs.

6.4 Data Collection Procedure

The survey was administered after the pilot deployment of MoodleQuest, allowing students to explore and interact with the system. Responses were collected electronically following system

use. No personally identifiable information was collected, and responses were treated anonymously to encourage honest and unbiased feedback.

6.5 Data Analysis Approach

Quantitative survey data were analyzed using descriptive statistics, including mean scores, standard deviations, and the percentage of positive responses, defined as ratings of 4 (Agree) or 5 (Strongly Agree). Survey items were grouped according to the three predefined constructs. Qualitative responses were analyzed through thematic grouping, where recurring patterns and common themes were identified based on student feedback.

7. RESULTS AND DISCUSSION

7.1 Overview of Survey Results

This section presents the results of the survey conducted to evaluate MoodleQuest as a gamified companion system for asynchronous learning. A total of 32 respondents completed 12 Likert-scale items, resulting in 384 quantitative responses. The overall mean score across all items was 4.16 with a standard deviation of 1.17, indicating a generally positive perception of the system. Approximately 64.8% of responses fell within the positive range (Agree to Strongly Agree), suggesting favorable acceptance among students.

Table 1. Overall Survey Statistics

Statistical Measure	Value
Total Respondents (N)	32
Total Item Responses	384
Grand Mean (M)	4.16
Standard Deviation (SD)	1.17
Percentage of Positive Responses (4-5)	64.8%

7.2 Results by Evaluation Category

Survey items were grouped into three evaluation categories: Perceived Engagement, Usability, and Learning Impact. Learning Impact received the highest overall ratings, with a mean score of **4.58** and **78.9%** positive responses. This indicates that students perceived MoodleQuest as having a strong influence on their study habits and learning outcomes. Perceived Engagement also yielded positive results, achieving a mean score of **4.38** and **71.1%** positive responses. Students reported increased motivation, enjoyment, and consistency when completing tasks using the system's gamification features. In contrast, Usability received a lower mean score of 3.53, with only **44.5%** positive responses. While some students found the system easy to learn and comfortable to use, others experienced difficulties related to navigation clarity and feature understanding. These results suggest that interface complexity and usability issues remain areas for improvement.

Table 2. Summary of Results by Evaluation Category

Category	Items	Mean	SD	% Pos
Perceived Engagement	4	4.38	0.98	71.1%
Usability	4	3.53	1.37	44.5%
Learning Impact	4	4.58	0.82	78.9%

7.3 Interpretation of Quantitative Findings

Analysis of individual survey items revealed consistently high ratings for learning-related outcomes. Items measuring improved study habits and overall learning impact received perfect mean scores, indicating unanimous agreement among respondents. Engagement-related items also scored highly, particularly those associated with motivation and enjoyment.

However, usability-related items showed greater variability in responses. While students reported ease of use and minimal need for assistance in some aspects, lower scores related to navigation and feature clarity highlight inconsistencies in the user experience. These findings suggest that while MoodleQuest is effective in motivating learning behavior, its interface design and information architecture require refinement to improve accessibility and ease of use.

7.4 Qualitative Feedback Analysis

Qualitative feedback reinforced the quantitative results. Students highlighted gamification features such as badges, leaderboards, virtual pets, and progress tracking as the most engaging aspects of MoodleQuest. Reported challenges included navigation difficulties, occasional performance issues, unclear reward criteria, limited customization, and accessibility concerns. Suggested improvements focused on enhanced pet customization, integrated study tools, additional social features, and improved analytics, aligning with the identified usability issues.

7.5 Discussion

The results demonstrate that MoodleQuest effectively enhances perceived engagement and learning impact in asynchronous Moodle courses. The strong ratings for learning outcomes and motivation suggest that transparent, teacher-guided gamification can positively influence student behavior without disrupting existing LMS workflows.

Despite these strengths, usability remains a key challenge. Lower usability ratings and recurring qualitative concerns indicate the need for interface simplification, clearer feature explanations, and improved navigation design. Addressing these issues is critical to ensuring that the system's motivational benefits are accessible to all users.

Overall, the findings support MoodleQuest's design goals while highlighting specific areas for refinement prior to broader deployment. The combination of quantitative and qualitative results provides a balanced evaluation of the system's effectiveness and usability.

8. CONCLUSION

MoodleQuest offers a practical and scalable approach to enhancing engagement in asynchronous learning by extending Moodle with transparent and teacher-guided gamification. By combining quests, experience points, badges, leaderboards, and behavioral analytics, the system encourages consistent study habits and provides instructors with clearer insights into learner progress. Early evaluations indicate that streak rewards and daily quests help students better understand their effort, while the integration approach maintains user trust by avoiding intrusive tracking and preserving Moodle's existing workflows.

Although challenges remain such as dependence on teacher validation and varying internet quality. The planned pilot studies aim to refine pacing, improve feedback loops, and assess the impact on student motivation. With further development in privacy-aware telemetry and objective-aligned recommendations, MoodleQuest has the potential to become a robust companion tool that strengthens learning outcomes in asynchronous environments.

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